

Bone Scintigraphy Retrieval Using SIFT-based Fly Local Sensitive Hashing

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Abstract—Bone scintigraphy is widely used in the diagnosis of bone diseases. With the expansion of bone scan database, there has been an increasingly demand for efficient technique to retrieve from the database. However, previous image retrieval methods mainly focused on nature images, and properties of nature images usually differ greatly from bone scintigraphy. In this paper, we apply a data-independent hashing method called Fly Local Sensitive Hashing (FLSH) for bone scan images retrieval. It's a computational strategies for solving approximate similarity search problem inspired by the olfactory system of fruit fly. To increase robustness for transformation and rotation, we also introduce SIFT features into retrieval procedure. SIFT is robust to various image transformation including scaling and rotating. A 128 dimension SIFT vector is used to represent the input images. We evaluate FLSH on our bone scan dataset containing 8091 images. The experimental results demonstrate that images retrieved by FLSH preserve more Euclidean similarity of query, and achieve higher precision compared to traditional LSH method. It indicates the effectiveness of SIFT-based FLSH in the area of bone scintigraphy retrieval.

Index Terms—Image retrieval, hashing method, bone scintigraphy, SIFT, Fly Local Sensitive Hashing (FLSH)

I. INTRODUCTION

Bone scintigraphy is widely used to detect metastases of bone such as tumors and cancers [1]. With the development of electronic medical technology, there have been researches focusing on segmentation [2]–[4], classification [5], [6] and enhancement [7] of pathological images. But few attention are paid on the area of retrieval, i.e., given a query scintigraphy, return from the database the top k images of which contents achieve highest similarity. It is also known as content-based image retrieval (CBIR), and can be regarded as an extension of the text-based search. Traditional methods address the problem of nearest neighbor retrieval via brutal search strategy, that is, compare the query image with images of the database one by one to seek the most relevant images. This strategy seems feasible when the scale of database is relatively small. Recent years, however, there has been a great increase in the amount of medical images available online, the increasing scale of the scintigraphy database makes it nearly impossible for brutal searching due to the huge consumption of computation and storage. It is a significant challenge to those computer vision applications that require high-efficiency similarity search.

Recently, enormous number of retrieval algorithms are proposed for alternation. Among these searching methods, hashing methods have been attracting increasingly attention for their higher scalability brought by efficient data representation and indexing mechanism. When solving the similarity searching problem, the idea of hashing is to construct a series of hash functions, which map the original data to hash codes. The core principle for choose hashing function is that, codes generated from similar images are still close to each other in the hamming space, and the binary codes will boost the efficiency of similarity retrieval.

Despite of the great results hashing methods have achieved, most of them focus on the retrieval of natural images. Fig. 1 exhibits two examples of bone scintigraphy, the properties of them are of great difference with those of natural images. First, the contents of bone scintigraphy usually involve tissues and organs against black background, which are basically similar for ordinary people. The main difference exists between two organs of the same kind but come from different people are caused by the variation in shapes and properties. Meanwhile, the perspectives of medical scintigraphy are fixed, i.e., all scan images are taken from a specific direction which is usually perpendicular to the objects. As a contrast, natural images are collected from various perspectives and images taken from objects of the same categories may have different appearance due to the variety of posture, scene and angle of view, which adds difficulty of retrieval. Second, scintigraphy is usually monochromatic or gray, thus contains less color information than natural images. Third, objects in scintigraphy are more likely to be garbled due to the diversity in morphological characteristics. The low imaging quality of X-ray also increases the difficulty of recognition.

In this paper, we exploit the Fly Local Sensitive Hashing (FLSH) for retrieval task of bone scintigraphy. FLSH is a data-independent hashing strategy found by Sanjoy et al. [8] when exploring the criteria for odor recognizing in the fruit fly olfactory circuit. We also introduce SIFT features into hashing procedure to increase the robustness of FLSH to various of noise and transformation. We apply FLSH to our scintigraphy database to study its effectiveness when dealing with bone scan images retrieval.

The structure of the paper is organized as follows. Section

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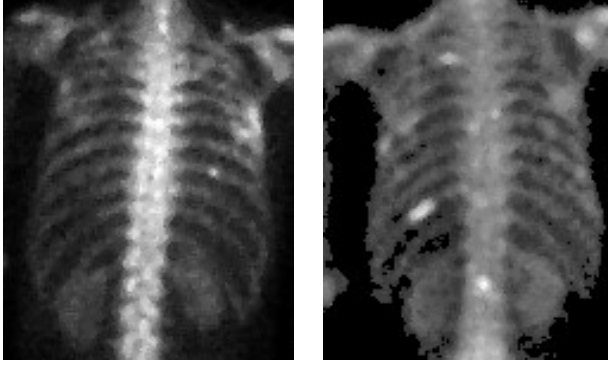


Fig. 1: Examples of bone scan image

II presents a brief overview of hashing methods. Section III gives a introduction to FLSH algorithm. Section IV introduce SIFT features into the bone scintigraphy retrieval procedure. Section V and VI provide experiment results and conclusions.

II. RELATED WORK OF HASHING METHODS

Hashing is similarity preserving encoding and can be regarded as a multi-binary classifier that assigns each image a label determined by its binary code. Existing hashing techniques can be broadly divided into two categories, data-independent and data-dependent methods. Data-independent methods construct hashing functions regardless the distribution of the original dataset or label information, while data-dependent methods generate hashing function with respect to the input dataset. One of the most popular data-independent hashing methods is Locality Sensitive Hashing (LSH) [9], which encodes images by projecting them onto a series of random generated hyperplanes. The disadvantages of LSH include low retrieval precision and high storage consume because it usually requires long bits for LSH to achieve considerable results. There are several improvements of LSH. Panigraphy proposed an entropy-based LSH [10] that generates several perturbing query objects that are close to the query object and queries them in addition to the query object to improve the precision of retrieval. Multi-probe LSH [11] adopts the same idea and makes a further step to look up brackets around the query data using a carefully derived probing sequence. There are also some generalization improvements in similarity and distance measures such as ℓ_p distance for $p \in (0, 2]$ [12], kernel similarity [13], [14] and Mahalanobis distance [15].

Data-dependent methods try to learn a series of hashing functions from the dataset to generate better codes. Iterative Quantization (ITQ) [16] is one of the representative unsupervised hashing methods, which finds the optimal projection matrix by iteratively applying rotating and projecting operation. One noteworthy problem when performing unsupervised methods is the existence of semantic gap due to the difference between Euclid distance and indeed similarity in semantic aspect [17]. Supervised methods are proposed to address this issue by incorporating supervised information into the hashing process. Most of supervised information is expressed as

pairwise labels of dissimilar or similar data pairs. Supervised Hashing with Kernels (KSH) [18] investigates the equivalence between a Hamming distance and a code inner product, leading to a efficient progress for minimizing the Hamming distance between similar objects and maximizing the Hamming distance between dissimilar pairs at the same time. Supervised Discrete Hashing (SDH) [19] uses the discrete cyclic coordinate descend method to transform the original problem into a convex optimization problem and optimizes the binary codes directly. Further improvements have been proposed to utilize deep neural network to learn the hashing process as well as image representation [20], [21].

Supervised methods have become dominant and achieve superior performance over the unsupervised hashing, but still data-independent and unsupervised methods hold their values for three reasons. First, the hashing process only requires data without supervised information, which are much more available comparing to labeled data. This property is greatly beneficial for medical image analysis because labelling requires much more professional knowledges, therefore supervised information is more difficult to obtain. Second, the training stage takes less time and computing resource. Especially, the data-independent methods can generate codes without the need for training stage, which makes it more easily to perform retrieval over large databases. Third, there is a great advantage over supervised method when the semantic gap of input images is neglectable, that is, the images with similar contents are close to each other in Euclidean distance as well. For example, same organs in bone scintigraphy are basically of the same shape lain in similar regions, and they are also close in color space due to the gray property of scan images, therefore same organs are more likely to be close to each other in Euclidean space. Finally, scenarios where multiple organs are included and overlapped with each other in one images also increase the difficulties of performing supervised methods. Existing supervised algorithms mainly focus on data with single main label, however, it's frequent to contain multiple organ labels in one scintigraphy, and there hasn't been an effective method to retrieve images containing multiply labels. Therefore, we currently focus only on hashing methods without the utilization of label information, such as FLSH.

III. FLY LOCAL SENSITIVE HASHING

Fly local sensitive hashing is a nearest neighbor searching strategy inspired by biological behavior of drosophila, it provides a distinctive view when solving similarity preserving encoding problem, and achieves performance which is 20 times faster than the commonly used LSH method. Fig. 2 shows the basic procedure of FLSH. Suppose we have a set of n data points $\{x_1, x_2, \dots, x_n\}$, $x_i \in \mathbb{R}^d$ which form the rows of the data matrix $X \in \mathbb{R}^{n \times d}$, the purpose of FLSH is to seek a group of hash functions $h : \mathbb{R}^d \mapsto \{1, -1\}^1$. The hash function is defined by $h_k(x) = \text{sgn}(xw_k)$, where w_k is one of the column vectors of a hash function coefficients and $\text{sgn}(v) = 1$ if $v \geq 0$ and 0 otherwise. We can also express the

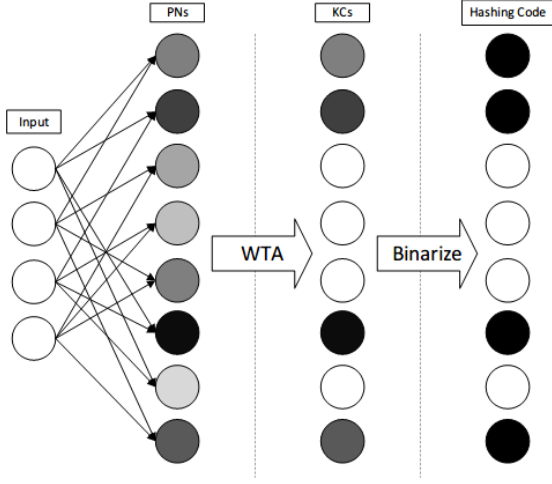


Fig. 2: Procedure of FLSH. PN stands for projection neuron and KC for kenyon cell, corresponding to the structure in fly olfactory.

whole process in matrix form as $B = \text{sgn}(XW)$, that is, input matrix X and output a binary code matrix $B \in \{-1, 1\}^{n \times m}$, where m denotes the code length and $W \in \mathbb{R}^{d \times m}$ is the hash matrix with columns w_k .

FLSH algorithm can be divided into three steps. In the first step, zero-centered technique is first apply to the dataset. It is a commonly preprocessing method used in many hashing algorithms and can be achieved by subtracting average value for each dimension, i.e., $\hat{x}_i = x_i - \sum_{i=1}^n x_i$. After that a dimension expansion operation maps the input vector x_i of d dimensions into hashes of m dimensions, i.e., $M = XW$, where $W \in \mathbb{R}^{d \times m}$ is the projection matrix. In the second step, a winner-take-all (WTA) mechanism is performed on each dimension of the projected data. For each column m_i of M , traverse the element of m_i , preserve only top 5% of the elements and set others to 0. During the third step, preserved

Algorithm 1: Fly Local Sensitive Hashing

Input : training data matrix $X = [x_1, \dots, x_n]^T \in \mathbb{R}^{n \times d}$,
the number of hash bits k , the sample rate η .
Output : binary code matrix $B \in \mathbb{R}^{n \times m}$.
 $x_i \leftarrow x_i - \sum_{k=1}^n x_k, i = 1, \dots, n$;
 $m \leftarrow 20k$;
randomly generate $W = [w_1, \dots, w_m] \in \mathbb{R}^{d \times m}$;
for $i := 1$ **to** m **do** *sparse binary projection*
 $n_{\text{sample}} \leftarrow \lfloor \eta d \rfloor$;
 random set n_{sample} elements of w_i to 1, while
 others set to 0;
end
 $B_0 = [b_1, \dots, b_n]^T \leftarrow XW$;
for $i := 1$ **to** n **do** *winner-take-all process*
 select top k elements of b_i and set these values to 1,
 while others set to 0, results in sparse binary vector
 b_i^* ;
end
return $B = [b_1^*, \dots, b_n^*]^T$.

elements are binarized, a $\text{sgn}(\cdot)$ function is applied element-wise, results in code matrix $B = \text{sgn}(M)$. Columns of B are sparse binary codes that satisfy $\sum_{j=1}^m b_{ij} = k, i = 1, \dots, n$. The number of elements that are set to 1 is equivalent to the length of the hash codes k , since only 5% of the elements are taken into account, we set $m = 20k$ to maintain enough valid code bits. The procedure of FLSH is described in Alg. 1.

Though both algorithms generate hash codes by random projections, FLSH differs from traditional LSH method in three aspects. First, the projection matrix W is sparse and binarized, only a small group (about 10%) of elements of w_i are nontrivial and set to 1, while others are 0. Traditional LSH method uses random but dense projection, and the elements of projection matrix are continuous. Second, the dimensionality of projecting space m is much higher than input space, i.e., $d \ll m$, which means the projection generates hash codes with a much longer length as compared to traditional LSH method, but the time complexity of computation remains the same due to the computing efficiency brought by sparsity of the projection [8]. Third, there is WTA mechanism to sparse the higher-dimensionality projection and obtain the final binary codes, which differs from the dense representation of traditional LSH method.

IV. SIFT-BASED FLY LOCAL SENSITIVE HASHING

The differences concluded in Section II highlight the advantages of data-independent hashing methods. However, there are several drawbacks for bone scintigraphy, such as the bad image quality and severe blur. The organs and tissues may also rotate and transform slightly, therefore we need a kind of feature that is invariant against transformation and robust against objects overlapping and sheltering. In this paper, we introduce Scale-Invariant Feature Transform (SIFT) [22] to the retrieval procedure, SIFT features are invariant to image scale and rotation, and also provide robustness when illumination are changing, which may appear frequently for medical scan images.

As described in [23], the procedure of SIFT algorithm can be divided into four stages. First, use Difference of Gaussian (DoG) to detect potential minimal points in the image as keypoints. Second, the value of normalized Laplacian function at each candidate keypoint is evaluated. If this value is smaller than some threshold, remove the keypoint. Third, according to local image gradient directions, each keypoint is assigned one or more main orientations. Finally, compute the keypoint descriptors according to the histogram statistics of the gradient orientation around the keypoints. If we divide the neighborhood of keypoints into 4×4 regions, and for

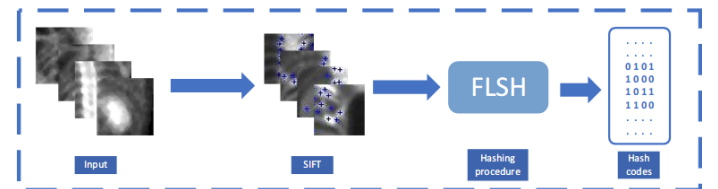


Fig. 3: Procedure of SIFT-based FLSH

each region the gradient information towards 8 directions is computed, then for each image we can obtain a 128 dimension feature vector. We then feed the SIFT features as input data to hashing method and generate binary codes. The whole procedure of the retrieval is present in Fig. 3.

V. EXPERIMENTS

In this section, three experiments are accomplished to demonstrate the effectiveness of FLSH. Our experiments were carried out on a desktop with a 3.40GHz Core CPU under MATLAB 2017 environment. The scan image database we used in experiments are collected from Department of Nuclear Medicine, Shanghai Renji Hospital. It contains 1414 bone scan images, the height and width of which ranges from 100 to 150 pixels. Four parts of organs, ribs, scapula, spine and kidney are involved. Fig. 1 exhibits two images randomly picked from the database. To set up retrieval database with uniform sample size, we randomly crop 8091 small patches from the original images with a fixed window size of 49×49 , and then represent them using SIFT features to present robustness to image scale and affine distortion [22] and reduce the dimensions of input space. Note that FLSH is a data-independent hashing method, which means data information is not involved during the whole procedure of the algorithm. In order to evaluate the performance of the method, we choose the Euclidean distance as the ground truth. This alternation is reasonable because the hashing methods involving random projection have been proved to have the ability to preserve similarity of Euclidean space [9], therefore the capacity for retrieval nearest neighbor of Euclidean space can reflect the performance of the algorithm. We choose LSH for comparison because it's the most popular data-independent hashing method.

A. Experiment I

In this experiment, we select a single image as query and use LSH and FLSH to retrieve the nearest neighbor. The query image is shown in Fig. 4a.

SIFT features are computed and fed into hashing methods, then for each hashing algorithm a hash code is generated. We compare the distance of the query codes with codes of dataset, and search the images with smallest Hamming distance. The retrieval results are shown in Fig. 4b and Fig. 4c. To measure the similarity of retrieved images, we compute the overlap rate between the query and retrieval image by

$$\frac{\sum_{x,y} \mathbf{1}[\mathbf{1}(I_q(x,y) > G_q) = \mathbf{1}(I_r(x,y) > G_r)]}{mn}, \quad (1)$$

where m, n are width and height of query image $I_q(x, y)$ and returned image $I_r(x, y)$, G denotes the average value of

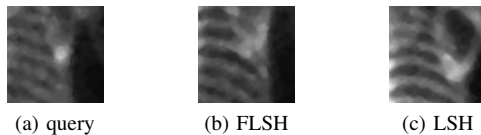


Fig. 4: Nearest neighbor retrieval using FLSH and LSH

TABLE I: Overlapping Rates of FLSH and LSH

Hashing Methods	FLSH	LSH
Overlapping rate	65.9424%	57.3486%

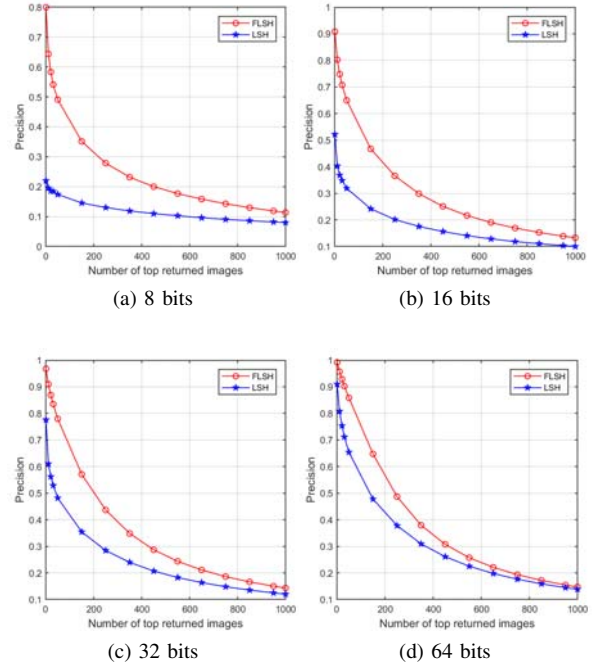


Fig. 5: Comparison of FLSH and LSH on bone scan image dataset using variant code lengths

images and $\mathbf{1}(x)$ denotes the indicator function that outputs 0 only if $x = 0$. The results shown in Table I indicates that the precision with SIFT-based FLSH is better than with LSH by around 8% increase.

B. Experiment II

In this experiment, we compare FLSH algorithm with LSH under different hash code lengths. We randomly select 1000 query inputs from the 8091, and for each query input, we compute the Euclidean distances of it with the feature vectors in database and select top 2% (161) of its nearest neighbors as ground truth sets. Then we successively use FLSH and LSH to encode the query, search top k of its nearest neighbors in hamming space to generate predict sets. The overlap between the ground truth sets and retrieval sets represents the number of true positive points. A precision curve is drawn by varying the number of top returned number images k . Results are shown in Fig. 5. It can be seen from the figure that FLSH outperforms traditional LSH method especially when the retrieval sets are small, both methods get improved and the gap between them are shrinking as the code size increases.

C. Experiment III

In this experiment, we compare the retrieval results of FLSH algorithm with traditional LSH method given a specific query

VII. ACKNOWLEDGMENT

This research is partly supported by NSFC, China (No: 61375048).

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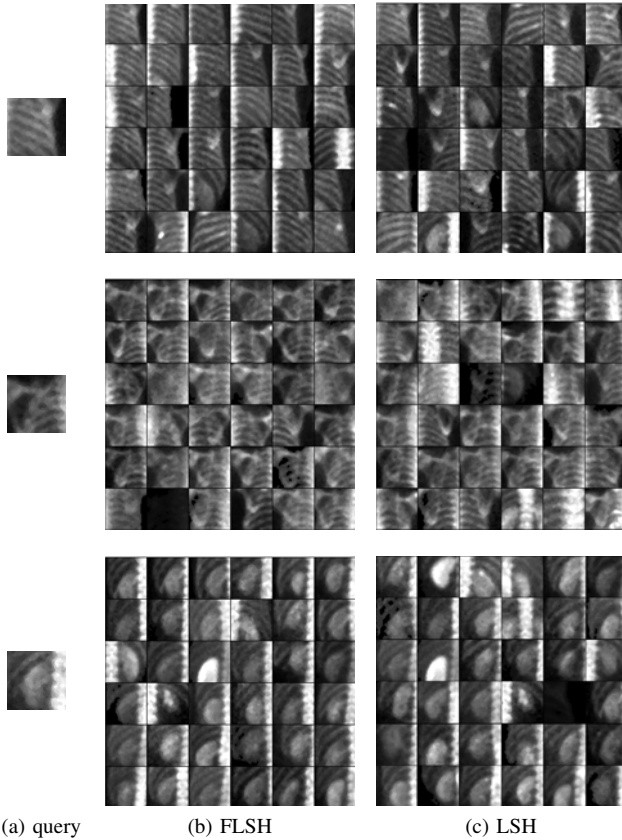


Fig. 6: Comparison of retrieval results between FLSH and LSH

image. For each query image we return top 36 images that are closest to it in Hamming distance. The results are shown in Fig. 6. We can see that results returned by FLSH express more morphological consistency, and the critical characters of the query image can be better preserved in the images retrieved by FLSH. For example, in the first group, i.e., the rib group, the rib of 4-th result returned by LSH presents a reverse orientation, while in the group retrieved by FLSH, the first reversed result appears at 24-th. For the second and third group, great difference can be capture from the first row of the retrieve results. Obviously, the results retrieved by FLSH show more consistence in shape and position compare to the query sample.

VI. CONCLUSION

In this paper, we combine Fly Local Sensitive Hashing with SIFT feature and exploit its performance on bone scintigraphy image retrieval. The algorithms are implemented and evaluated in MATLAB environment and the dataset used for evaluation are collected from Renji hospital, Shanghai. Experiments show that FLSH is capable of bone scan image retrieval and can achieve considerable result comparing with traditional LSH method. In future work, we will extend our focus to the utilization of data information when performing FLSH to enhance the performance.

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